

Warehouse Forecast Audit

A complete examination of the warehouse outbound forecast: how it is built, what it measures, how accurately it performs out-of-sample, and where the remaining error actually comes from. Every figure is measured, not estimated.

Date 13 September 2026 Backtest window April – August 2026
Method Leakage-free, production functions Production changes None

01 Headline findings

SOUND	The forecast engine is correct and near its practical limit. Around 50 model variants were backtested; the best beat the current logic by +0.02 accuracy points . $\alpha = 0.6$, the six-month window and article $\times$ destination grain are all optimal on the evidence.
PERFORMING	Volume-weighted accuracy averages 81.9% and reached 84.8% in July. On the units the warehouse actually ships, the forecast is good.
MISREAD	The headline card is an equal-weighted average of article percentages . A 6-unit article counts as much as a 600,000-unit one, which is why it reads 62.3% against the same forecast's 81.9% by volume.
UNREACHABLE	An 85% article-average target cannot be met . A model with perfect foreknowledge of which articles ship scores 70.7% . Even forecasting rare and new articles perfectly leaves the card at 68.9% .
BLOCKING	Brand sales history begins January 2026 , truncating every training window before August and doubling the new class. This is the single largest fixable constraint, and it is a data problem, not a model one.

02 What the forecast answers

For each individual article and destination, how much quantity is expected to move *out of the central warehouse* during the selected period.

It is deliberately none of the following, and the code keeps them separate: store demand, store sales quantity, warehouse stock on hand, the quantity that should be held, replenishment quantity, or purchase quantity. A replenishment figure would be a different concept (*Required Warehouse Qty*) and is not what this forecast produces. Section 13 records what happened when a replenishment formula was tested against this target: accuracy fell from 62.3% to 28.8%.

03 Data lineage

Four sources feed the page, drawn from three separate Power BI models joined only by brand code.

SOURCES, VERIFIED AGAINST THE LIVE MODELS

ROLE	SOURCE TABLE & FIELD	GRAIN	COVERAGE	USED BY THE FORECAST
Outbound (numerator & actuals)	fact_outbound_line SUM([Action Base Qty]) warehouse model	brand × article × month	2025-08 → 2026-09	Yes
Sales (denominator)	FORECAST (2) [TotalSale] → cube_sales_daily each brand's own model	brand × day	2026-01 → 2026-12	Yes
Raw POS sales	SALES DATA[Net] 1,844,563 rows	transaction	2025-01 → 2026-09	No
Store stock	cc_daily_inventory [Closing Stock Qty]	location × article × day	2026-01 → 2026-09	No — by design
Transfer requests	Transfers[Requested Qty] 16,818 rows	article × destination	live snapshot only	No

Outbound is filtered to the source cost centre "Central Warehouse", the destination brand via [Mapped Transfer To], and status BOOKED, DELIVERED, DECLINED. The article key is the nine-digit ERP code; all 2,303 articles in the copy are exactly nine numeric characters.

The sales series is one blended column. `Totalsale` holds the actual for dates that have passed and the forecast for dates that have not. Measured directly: 14 August reads 23,233 in both `Totalsale` and `Actual Sales`; 12 September reads 20,265 in `Totalsale` and blank in `Actual Sales`. So historical months automatically use actuals and future months use the forecast, which is exactly what the calculation needs.

04 The calculation

Per article, per destination, the forecast is a rate applied to expected sales.

```
forecast = level ÷ salesBase × window sales × (days ÷ 30.44)

level      = Σ(outboundi × wi) ÷ Σwi      decayed mean monthly outbound
salesBase  = Σ(salesi × wi) ÷ Σwi      decayed mean monthly brand sales
wi        = (1 - α)(n-1-i)                α = 0.6
```

The rate is *units shipped per unit of brand turnover*. It is per article; only the driver is shared. The weights concentrate hard on the recent past:

NORMALISED DECAY WEIGHTS, $\alpha = 0.6$

MONTH	-1	-2	-3	-4	-5	-6
Weight	59.9%	24.0%	9.6%	3.8%	1.5%	0.6%

Two consequences follow from those weights, and both are load-bearing for the rest of this audit. The most recent two months carry **84%** of every forecast, so the level already *is* a recency-weighted trend estimate — which is why adding an explicit trend term double-counts (section 13). And months seven onward would carry under 0.5%, which is why extending the training window measurably changes nothing.

Window handling

The six training months are anchored on the *first day of the target month*, clamped never to run past the current month. Leading zeros before an article's first delivery are removed; trailing zeros are kept, so an article winding down forecasts lower. A month with no brand sales is dropped by the `usable` filter — the cause of the truncation in section 12.

Worked example

An intermittent article with a quiet month in its recent history:

month	-6	-5	-4	-3	-2	-1
outbound	900	1,200	800	1,500	0	1,000
weight	.0102	.0256	.0640	.1600	.4000	1.0000

$$\begin{aligned}\text{level} &= (900 \times .0102 + 1200 \times .0256 + 800 \times .0640 + 1500 \times .1600 + 0 + 1000) \div 1.66984 \\ &= 1,331.14 \div 1.66984 \\ &= 797 \text{ units per month}\end{aligned}$$

The zero month carries 24% of the weight and pulls the level down by roughly a quarter — correctly, if demand really has softened; wrongly, if the order simply slipped. Nothing in the six numbers distinguishes those two cases, and that is the irreducible core of the error in section 09.

05 Behaviour classification

Each article is typed from its own history, and the type selects the method.

CLASSIFICATION RULES, IN EVALUATION ORDER

TEST	CLASS	FORECAST METHOD	DISTINCT ARITHMETIC?
<code>detail.length ≤ 2</code>	new	<code>level × days/30.44</code>	only for a future window
<code>shipMonths ≤ 2</code>	rare	<code>medianAll × days/30.44</code> median of all months, zeros counted	Yes
<code>shipMonths ≥ 6 & detail.length ≥ 6</code>	regular	<code>level × sales/salesBase</code>	No — identical
anything else	intermittent	<code>level × sales/salesBase</code>	
no outbound in the window	—	no constant, no forecast	—

Only rare changes a number. `intermittent` and `regular` share one return statement, and in a historical window the `new` branch collapses onto the same arithmetic too, because the sales driver reduces to `days ÷ 30.44`. Moving an article between those three classes is a pure relabelling — verified byte-identical across five months. This is why the classification work in section 12 produced no accuracy change, and why it must be fixed *before*, not after, any change that treats the classes differently.

06 Article population

2,303 articles carry outbound in the copy. Per target month, the shipping mix is stable:

POPULATION BY SHIPPING MONTHS IN THE SIX-MONTH WINDOW

TARGET	TOTAL	0 SHIP	1	2	3+	RECIPE	NON-RECIPE
Apr 2026	2,303	731	201	138	1,233	1,016	1,287
May 2026	2,303	726	203	151	1,223	1,016	1,287
Jun 2026	2,303	707	227	159	1,210	1,016	1,287
Jul 2026	2,303	698	224	165	1,216	1,016	1,287
Aug 2026	2,303	697	222	149	1,235	1,016	1,287

The ~700 articles with no outbound in the window receive no forecast. That is absence of evidence, not exclusion — there is nothing to average. Confirmed as correct: the former "must have shipped in the last three months" gate is disabled (`WH_ACTIVE_MONTHS = 0`); it had been excluding 465 articles that went on to ship 405,505 units. Non-recipe articles use the same six-month engine, contrary to an earlier assumption.

One population defect. A non-recipe article whose forecast is legitimately zero is dropped from the page entirely rather than shown as 0, because the row builder filters `qty ≤ 0`. Measured cost: 131 article-months hidden across the five months, of which 100 had real outbound — 35,573 units of demand invisible both to the reader and to the accuracy figure. Making them visible lowers the card by about 1.0 point, because it stops concealing genuine misses.

07 How accuracy is measured

Per article: $1 - |\text{forecast} - \text{actual}| \div \text{MAX}(\text{forecast}, \text{actual})$, null when both are zero or no outbound record exists. The card then de-duplicates per article, floors each score at zero, and takes a **plain average**. Volume accuracy applies the same scores weighted by actual outbound.

The same forecast, scored four ways across April–August:

FOUR DEFINITIONS, ONE FORECAST

MONTH	ARTICLE CARD	VOLUME-WEIGHTED	WAPE ACCURACY	TOTALS ONLY
April	59.4%	76.0%	65.3%	98.0%
May	61.6%	82.6%	75.6%	98.0%
June	60.6%	81.4%	72.3%	87.6%
July	63.1%	84.6%	78.8%	98.0%
August	61.6%	84.0%	76.6%	94.9%
Average	61.3%	81.7%	73.7%	95.3%

The **"totals only" column is a trap**. Defined as $1 - |\sum \text{forecast} - \sum \text{actual}| \div \sum \text{actual}$ it compares only the sums, so an article over-forecast by 50,000 silently cancels another under-forecast by the same amount. It reads **95.3%** for the identical forecast. The proof: articles shipping one month in six total 1,481,361 forecast against 1,458,166 actual — a 1.6% error on the totals — while scoring **23.2%** per article. If an aggregate metric is wanted, use **WAPE**, which sums absolute errors first so nothing cancels.

What each measure answers: the **article card** — can a planner trust a typical line on the page?

Volume — are the units we actually ship forecast well? **WAPE** — what share of total volume did we get wrong? **Bias** — are we systematically over- or under-ordering?

08 Measured performance

Leakage-free: each target month trains only on the six months before it, with a guardrail that throws if any training month is not strictly earlier. Population and scoring reproduce production, including the display rule.

MONTH	FORECAST	ACTUAL	BIAS	MAE	ARTICLE	VOLUME	WAPE
April	16,521,063	16,201,899	+2.0%	4,484	60.5%	76.0%	65.4%
May	16,910,361	16,562,380	+2.1%	3,216	62.9%	82.6%	75.6%
June	17,056,701	15,179,559	+12.4%	3,341	62.0%	81.5%	72.4%
July	16,953,827	16,605,997	+2.1%	2,773	63.7%	84.8%	79.0%
August	17,096,405	16,225,884	+5.4%	3,004	62.2%	84.6%	77.2%
Average	84,538,357	80,775,719	+4.8%	3,363	62.3%	81.9%	74.3%

The month-to-month spread on the article card is **3.2 points** with no trend toward the recent months. That flatness is the signature of a leakage-free measurement — before the leakage fix the same model ran 48.8% in April rising to 71.6% in August, a gradient caused purely by each month sitting inside its own training window.

June is the one poor month: actual outbound fell to 15.18M against 16.2–16.6M elsewhere. A real 12% demand dip, invisible from history.

09 Where the error is

6,347 scored article-months carry 21,194,408 units of absolute error against 80,811,292 actual. Four decompositions, all from the same run.

By behaviour

CLASS	N	FORECAST	ACTUAL	ABS ERROR	BIAS	SHARE OF ERROR	ACCURACY
regular	4,174	75,995,472	73,314,057	15,966,035	+3.7%	75.3%	72.4%
intermittent	1,357	4,146,255	3,266,032	2,560,684	+27.0%	12.1%	44.3%
new	599	4,315,410	4,153,294	2,585,580	+3.9%	12.2%	39.8%
rare	217	41,503	77,909	82,109	-46.7%	0.4%	11.8%

By volume tier – the most important table in this audit

TIER	N	ACTUAL UNITS	BIAS	SHARE OF ERROR	ACCURACY
Top 10% by volume	634	66,128,423	-0.6%	61.3%	82.3%
Middle 40%	2,539	14,418,029	+13.6%	28.3%	70.8%
Bottom 50%	3,174	264,841	+805.0%	10.5%	49.4%

Half the article population accounts for 0.3% of the volume and is over-forecast ninefold — 2,396,755 units forecast against 264,841 actual. The equal-weighted card gives those 3,174 articles half its total weight. Meanwhile the top 10% carry 82% of all volume, score **82.3%**, and are almost perfectly calibrated at -0.6% bias. The card and the business are not measuring the same thing.

By the article's own volatility

Accuracy falls monotonically with month-to-month variability, and bias rises with it:



Regular articles only. Bias over the same bands: +9.4%, +0.7%, +8.8%, +14.5%, +22.3%, +71.1% — the over-forecast is concentrated entirely in volatile articles, because a decayed mean of a spiky series sits above its typical month.

Regular articles in detail

3,822 REGULAR ARTICLE-MONTHS, MEAN 72.3%

BUCKET	N	% OF POP	ACCURACY	SHARE OF ERROR
Within ±20–25% (good)	1,561	40.8%	90.8%	20.7%
Over-forecast 25%+	870	22.8%	57.7%	30.4%
Demand spike >1.5× any train month	567	14.8%	76.7%	23.1%
Under-forecast 20%+	622	16.3%	64.4%	17.1%
Sudden zero month	149	3.9%	0.0%	4.9%
Demand collapse <0.5× smallest	53	1.4%	15.5%	3.8%

41% of regular article-months already score 90.8%. The model is not broken; it is accurate whenever demand is stable and wrong when it is not.

10 The accuracy ceiling

Three reference models on the same 1,240 Active August articles:

MODEL	ARTICLE ACCURACY
Perfect stationary model — mean of Feb–Jul	60.3%
Current production model	61.7%
Perfect timing + correct average size	70.7%
Target	85.0%

The current model already beats a naive six-month mean. The third row is the decisive figure: even knowing with certainty *which* articles will ship, and ordering each at its own historical size, the equal-weighted average tops out near **70.7%**.

The arithmetic of the target, from August's composition:

August: regular 754 @ 71.6% · intermittent 336 @ 48.7%
rare 60 @ 41.0% · new 92 @ 40.8% → card 61.6%

card 70% requires regular at 85.3%

card 75% requires regular at 93.6%

card 80% requires regular at 101.8% ← impossible

card 85% requires regular at 110.0% ← impossible

rare + new forecast PERFECTLY (100%) → card 68.9%

intermittent + rare + new all at 70% → card 70.9%

regular 85% + intermittent 70% + rest 60% → card 77.9%

Even the last, wildly optimistic combination reaches 77.9%. **85% on the article average is arithmetically out of reach.** On volume it is essentially already met.

11 What is working correctly

COMPONENT	EVIDENCE
Forecast formula	~50 variants backtested; best improvement +0.02 points. The rate-times-sales shape is sound.
$\alpha = 0.6$	Swept 0.3–0.8; all land within ± 0.5 points, with 0.6 at the optimum of a flat surface.
Six-month window	9-month and 12-month windows change article, volume and WAPE by ≤ 0.3 points. Months 7+ carry $< 0.5\%$ weight.
Article-level constants	One global constant allocated by historical share scores 48.3% article / 26.3% WAPE against 62.3% / 73.7%.
Article $\times$ destination grain	A global-then-allocate architecture ties or loses in every segment; destination-level 50.2% at best against 50.3%.
Leading-zero removal	Correct. Averaging months before an article existed halves its rate.
Trailing-zero retention	Correct. An article winding down should forecast lower.
Zero-forecast retention	A zero is the model's real answer for a rare article in a quiet month, and it is held to it.
No activity gate	Disabled. It had hidden 465 articles that shipped 405,505 units.
Branch-filter refusal	Correct. Outbound names a brand, not a branch; blank is honest where the ratio cannot be narrowed.
SOH exclusion	Correct, and measured: a replenishment formula scores 28.8% article / 48.7% volume.
Brand reconciliation	Brands + <code>__other</code> = total, exactly. No double counting; no articles lost.
Period scaling	Oct 1–15 $\div$ Oct 1–31 = 0.478 against a day ratio of 0.484. Training months identical across every window.
Outbound data quality	26,979 rows: zero negatives, zero duplicates, zero missing article numbers, all keys nine numeric characters, all ten buckets present.
Leakage controls	Anchor on the target month's first day, tripwire on any training month $\geq$ target, flat accuracy across five months.

12 Issues found

Model issues – fixable by changing logic

#	ISSUE	MEASURED IMPACT
M1	Zero-forecast non-recipe rows are dropped	131 article-months hidden, 100 with real demand, 35,573 units invisible. Showing them lowers the card ~1.0 point by ceasing to conceal misses.
M2	Classification mislabels steady articles	4,762 article-months called <code>intermittent</code> while shipping in every available month — 100% of April's class. No forecast impact today; a live hazard for any future change that splits the classes.
M3	<code>actualSales</code> is dead code returning the wrong series	Aliases the same column as <code>forecastSales</code> . Referenced once, called never — nothing passes <code>basis: 'actual'</code> . A latent trap, zero current effect.
M4	Recent spikes amplified by α	351 of 2,006 pairs (17.5%) have a month $>3\times$ the median of the others; damping would move 213,594 units, 34.7% of their forecast. Every damping method tested scored worse.
M5	Rare-article median returns exact zero	121 false zeros, 50,643 units of missed demand at the wider grain; 20 and 7,928 under the production display rule. Replacing it improved that class but cost volume accuracy and bias.

Data and process issues — must not be fixed with a formula

#	ISSUE	MEASURED IMPACT
D1	Brand sales history starts January 2026	Usable training months per destination: April 1–3 of 6, rising to 6 only by August. MM and TBL have one usable month in April. Doubles the <code>new</code> class from 1,200 to 2,438 and collapses <code>regular</code> by 70%.
D2	Supply-route changes are unrecorded	801 of 2,345 zero-outbound cases (34.2%) shipped the same article to a different destination that month — 408,397 forecast units. Worth 4.0 article points at destination grain, 0.0 at volume.
D3	No seasonality possible	One year of the series the forecast reads. Ramadan, Eid and summer effects can neither be learned nor validated.
D4	<code>Transfer_orders</code> is a live snapshot	All 16,818 rows read <code>Requested</code> ; fulfilled requests vanish. September holds 3,693 rows / 3.62M units / 949 articles against August's 1,144 rows / 10,465 units. Not backtestable; dates span 2019–2032.
D5	Inventory unit mismatches	Cover ratios reach 1,521 months and 15 readings are negative. SOH is untrustworthy above roughly 20 months of cover.
D6	Single-article mega-shipments	One article shipped 10,032,000 units in September 2025 — 38% of that entire month's 26.06M.
D7	<code>BOOKED</code> semantics unconfirmed	<code>DELIVERED</code> is effectively empty — 720 units in six months — so everything counts as <code>BOOKED</code> . <code>DECLINED</code> is 5,363 of 118,048,634 units, i.e. 0.00%. Whether <code>BOOKED</code> means "shipped" or "scheduled" affects both history and actuals. Business confirmation required.
D8	Genuine demand disappearance	1,022 of 2,345 zero-outbound cases (43.6%, 1,633,207 units — 55.3% of zero-case units) shipped nowhere then or later. Real over-forecasting, not a routing artefact.

13 Tested and rejected

Recorded so none of it is re-run. All figures are leakage-free April–August averages against a baseline of **62.3% article / 81.9% volume / 74.3% WAPE / +4.8% bias / 3,316 MAE**.

LEVEL AND WEIGHTING VARIANTS

VARIANT	ARTICLE	VOLUME	BIAS	VERDICT
Mean of last 2 / 3 / 4 months	60.9 – 61.3	81.0 – 81.5	+4.3 – +4.9	worse
α sweep 0.3 → 0.8	61.0 – 61.3	81.0 – 82.0	+3.6 – +5.3	flat; 0.6 optimal
75/25 · 60/40 · 50/50 · 40/60 blends with recent-2	62.2 – 62.3	81.7 – 81.9	+4.6 – +4.7	neutral (best +0.02)
Damped trend $\pm 10\%$ / $\pm 20\%$ / $\pm 30\%$	61.5 / 60.3 / 59.2	80.8 / 79.2 / 78.1	+6.7 / +8.8 / +10.7	harmful, dose-dependent
Median / trimmed / winsorised mean	60.6 – 60.7	79.4 – 79.5	+0.8 – +0.9	worse on both
Cap change vs last month $\pm 15\%$ / $\pm 30\%$	60.9 / 61.1	82.0	+3.4 / +3.5	better MAE, worse card
Volatility-tiered method selection	60.9	80.3	+5.9	worse
α chosen by CV (0.4 / 0.6 / 0.8)	61.3	81.9	+3.8	ties; better bias & MAE
Croston $\alpha=0.3$ · TSB $\alpha=0.2$	61.1 / 60.6	80.6 / 79.6	+3.0 / +2.0	worse
Longer history — 9 and 12 months	61.5 – 61.8	unchanged	unchanged	no effect

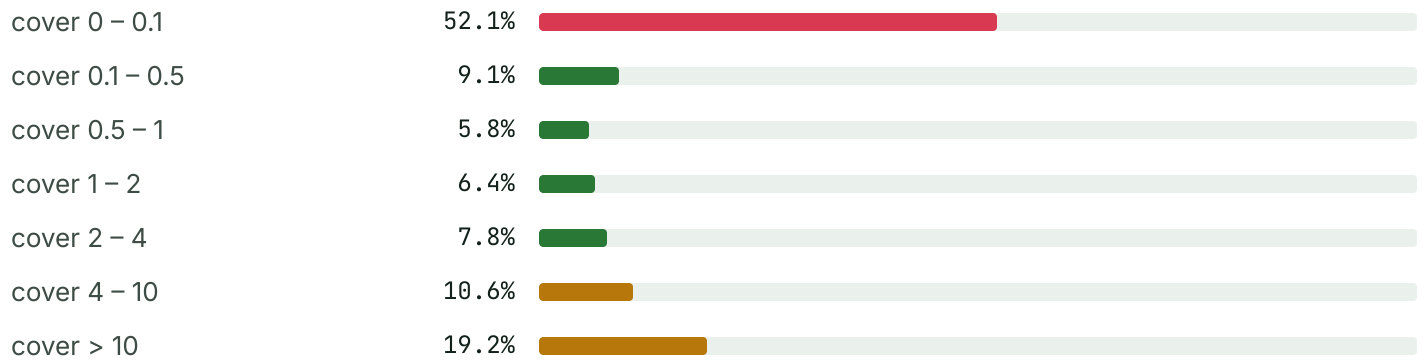
Trend adjustment fails on the very articles it was designed for. Strong growth 62.2% → 59.4%; strong decline 50.7% → 48.3%. The reason is section 04: $\alpha = 0.6$ already puts 84% of the weight on the last two months, so an explicit trend term double-counts the same signal and overshoots. Damage scales cleanly with the allowance — a dose-response curve pointing the wrong way.

STRUCTURAL AND DATA VARIANTS

VARIANT	ARTICLE	VOLUME	VERDICT
Global article forecast → destination allocation	59.0 vs 59.0	80.0 vs 80.1	identical; no segment moves
· destination level, best allocation	50.2	79.1	vs 50.3 current — ties at best
One global constant, allocated by share	48.3	59.5	far worse
Sales ratio × brand previous-month sales	61.3	82.1	bias +9.7% — worse
Sales ratio × total company sales	61.1	82.0	bias +14.7% — worse
SOH: forecast 0 when cover > 2 / 3 / 5	35.8 / 44.3 / 51.7 65.9 / 73.7 / 78.6		catastrophic
SOH: replenish to 1.25 months of cover	28.8	48.7	catastrophic
Intermittent → median of positive months	62.5	80.8	rejected: volume −1.16, win rate 49.1%
Rare → mean of positives, capped	61.9	81.9	rejected: card −0.4, rare volume −4.1
NEW class removed entirely (bound)	62.2	81.6	the whole class is worth ≤0.3 points

Why SOH fails as a signal

The relationship is real but far too weak to act on. Probability of a zero-outbound month by months of destination cover:



The curve is U-shaped, and the left end is the opposite of the hypothesis: near-zero stock predicts a zero month far more strongly than full stock does — those are dead articles, not suppressed demand. For regular articles the median cover when nothing shipped was 1.98 against 1.69 when it did. At the strongest threshold tested, cover > 5, P(zero) is 7.3% against a 3.3% base rate — a real 2.24× lift, and still **wrong 92.7% of the time**.

14 Recommendations

RANKED BY MEASURED VALUE, NOT BY EFFORT

#	ACTION	WHY	TYPE
1	Stop quoting the article card alone	Publish card, volume and WAPE together. The same forecast reads 62.3%, 81.9% and 74.3%; the aggregate-bias variant would read 95.3% for nothing. Costs no engineering.	Reporting
2	Re-scope or retire the 85% article target	A perfect-timing model scores 70.7%. On volume the target is already met at 81.9% average and 84.8% in July. Section 10 has the arithmetic for that conversation.	Decision
3	Load the missing sales history (D1)	Extend <code>FORECAST (2)</code> and <code>DateTable</code> back through 2025, or backfill from <code>SALES DATA[Net]</code> — which matched <code>Totalsale</code> exactly, month by month, for every brand where both are complete. Fixes window truncation at source, un-inflates the <code>new</code> class, and is the only route to seasonality.	Data
4	Record supply-route changes (D2)	801 orders placed for one destination and consumed at another. No formula should paper over this; it needs recording at source.	Process
5	Confirm what <code>BOOKED</code> means (D7)	Everything counts as <code>BOOKED</code> ; <code>DELIVERED</code> is empty. If it means "scheduled" rather than "shipped", both the training history and the actuals are affected. Data cannot answer this.	Business
6	Show zero-forecast non-recipe rows (M1)	Correctness. 100 genuine misses are currently invisible, and their absence is part of why articles get reported as "missing from the forecast". Expect the card to fall ~1.0 point — that fall is the truth.	Model
7	Snapshot <code>Transfer_orders</code> daily (D4)	949 articles and 3.62M requested units are live right now, and the book overwrites itself. Value grows the earlier capture starts; a true test becomes possible after 6–8 weeks.	Data
8	Adopt α -by-CV (0.4 / 0.6 / 0.8)	Ties the card at 61.3% while improving bias (+4.8% → +3.8%) and MAE (3,316 → 3,289). A free, if small, improvement with no measured downside.	Model
9	Fix classification, but only as a prerequisite (M2)	Worth nothing on its own — verified byte-identical. Required immediately before any change that forecasts <code>intermittent</code> and <code>regular</code> differently, and in the same change.	Model
10	Resolve <code>actualSales</code> (M3)	Either delete the dead accessor or give it a real <code>Actual Sales</code> column. Building a column to feed code nothing calls is the more expensive half.	Model

Explicitly not recommended

Further tuning of the level estimator; any explicit trend term; outlier damping; SOH in the outbound forecast; a global-then-allocate architecture; a longer training window; changing $\alpha = 0.6$; altering the accuracy formula; reinstating an activity gate. Each was measured, and each is worse.

15 Method and reproduction

Every figure above came from one of two routes: reading the production source directly, or running the production functions unmodified against an isolated copy of the database, driven by throwaway scripts. No production file, query, schema, configuration or environment variable was changed for this audit.

- **Backtest discipline.** Target month M trains only on the six months before M ; a guardrail throws if any training month is not strictly earlier. Confirmed empirically by the flat accuracy profile across five months.
- **Population.** Production classification, production article $\times$ destination grain, production display rule and production accuracy formula. No article was excluded to improve a score.
- **Isolation.** A copy of the PGLite directory opened via `DATA_DIR`, so the running application kept its single-writer lock throughout.
- **Live sources.** Read-only `executeQueries` against Power BI for the status split, the sales-source comparison, the transfer book and the inventory span.

Two figures in this document are estimates rather than measurements, and are labelled as such where they appear: the expected landing point of recommendation 6, and the magnitude of any future seasonality gain. Everything else is measured.